LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 3: Project: Survey Analyser

Module 1 | 50 minutes | Project: data app | Grades 6–8 · ages 11–14

Objective: Plan, build and test an app that collects survey answers into a list and reports statistics.

Success criteria

- My app collects answers into a list.
- It reports correct statistics.
- I followed data-ethics rules.

Key vocabulary

- **data set**: A collection of values.
- **anonymous**: Not linked to who gave it.
- **statistic**: A number that summarises data.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Hands-up survey (5 min): collect a quick class tally and compute the average on the board.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Is this survey question appropriate?. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Survey Analyser.

Unplugged warm-up

Hands-up survey (5 min): collect a quick class tally and compute the average on the board.

Teaching the idea

Integrates list building from input, traversal, accumulators and extreme-value scans, plus data ethics (minimisation, anonymity, consent).

Analogy: A census: collect, count, summarise.

Live demo: Run the analyser with 3 known values and check each statistic.

Questions to ask

Ask	Listen for
Why test with known data?	You can check the answers.
What makes a survey question ethical?	Anonymous, necessary, non-sensitive.
What insight did you add?	E.g. how many above average.

Differentiation

Support
Provide the collection loop; learners write the analysis.

Stretch
Count how many responses are above the average.

Common misconceptions

Misconception	How to fix it
More data is always better	Collect only what you need.
The program is right if it runs	Check with known data.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Data collection	None	Fixed data only	Collects input into a list	Flexible count with validation
Analysis	None	Total only	Average, max and min correct	Extra insight (e.g. above average)
Code quality	Hard to follow	Works	Clear loops and variable names	Elegant, reusable structure
Testing	None	Ran once	Tested with known data	Several test sets including edge cases
Ethics & reflection	Sensitive data	Unclear	Anonymous and appropriate	Explains data choices clearly

Exit ticket
One rule for collecting data responsibly.

Homework
Run your survey with family (anonymously) and share one finding.

Answer key — Kitabash activities

1. Is this survey question appropriate? (perform · categorize)

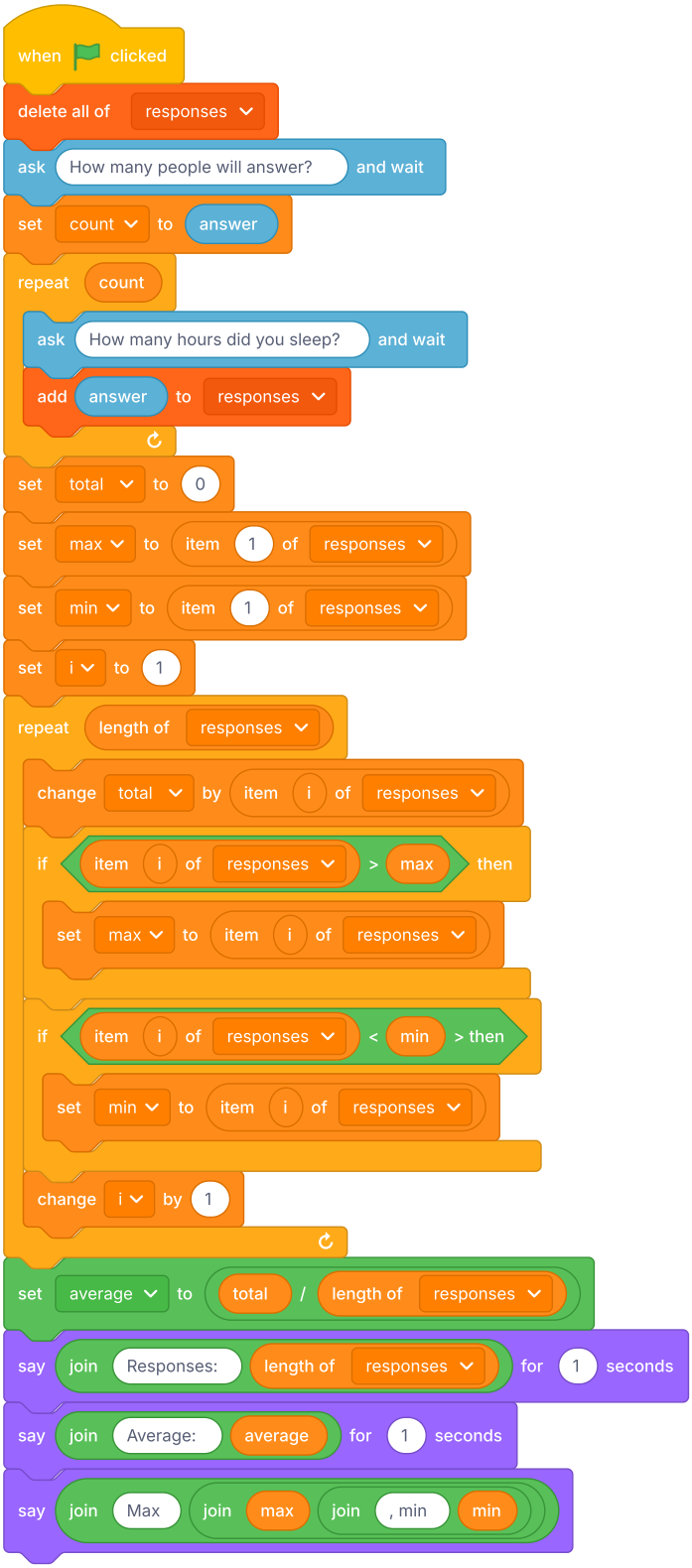
Good question: How many hours did you sleep last night?, How many books did you read this month?, How many minutes do you walk to school?
Needs changing: What is your full name and home address?, How much money do your parents earn?

2. Project: Survey Analyser (probe · code lab)

Goals checked: Collects answers · Traverses the list · Calculates · Correct average

Example project:

Analyst



Answer key — worksheet

1. **Key words:** data set = A collection of values. · anonymous = Not linked to who gave it. · statistic = A number that summarises data.
2. **Is this survey question appropriate?:** Good question: How many hours did you sleep last night?, How many books did you read this month?, How many minutes do you walk to school? · Needs changing: What is your full name and home address?, How much money do your parents earn?
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.